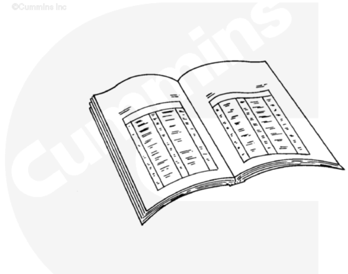
Trainings

General Information



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Before any cylinder block inspection, cleaning, or repair, determine the assignable cause of the malfunction.

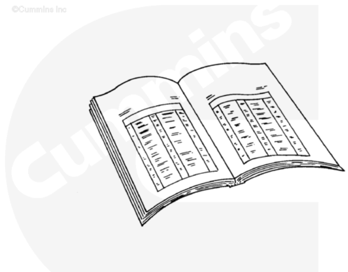
Knowing the assignable cause of the malfunction and using the appropriate cleaning procedure will lead to a high quality repair.

Once the assignable cause of the malfunction is determined, the appropriate cylinder block cleaning procedure from the table below can be followed.

Note : If replacing the cylinder block with new or using a previously stored cylinder block, make sure to clean any oil/rust residue preventative solvent from the cylinder bores, main bearing bores, and gasket sealing areas prior to assembly.

Note : Replace the lubricating oil cooler if the engine has had a debris-generating malfunction.

Assignable Cause	Cleaning Process	Additional Instructions
Debris/Shavings Contamination	Cleaning tank or non-cleaning tank	Replace the lubricating oil cooler if the engine has had a debris-generating malfunction



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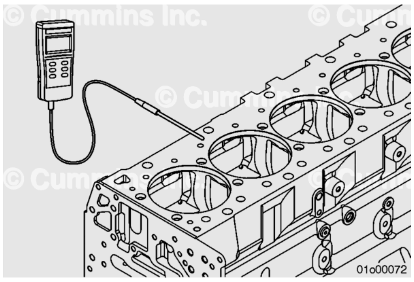
Assignable Cause	Cleaning Process	Additional Instructions
Dust Out/Air Intake Contamination (foreign object debris)	See note in next column	If a foreign object debris malfunction caused a secondary malfunction that resulted in debris/shavings contamination of the lubricating oil system, the debris/shavings malfunction cleaning process **must** be followed.
Fuel Dilution	See note in next column	If a fuel dilution malfunction caused a secondary malfunction that resulted in debris/shavings contamination of the lubricating oil system, the debris/shavings malfunction cleaning process **must** be followed.
Magnetism of ferrous component due to electrical current	Cleaning tank or non-cleaning tank	Cylinder block and non-ferrous components **must** be demagnetized or replaced. Refer to the magnetism inspection in the General Information section of this manual.

Assignable Cause	Cleaning Process	Additional Instructions
Oxidation (scaled) from water contamination	Cleaning tank or non-cleaning tank	

If evidence of ferrous material clinging to the cylinder block is found, the cylinder block **must** be checked for magnetism. An indication of magnetism is ferrous shavings clinging to the cylinder block during cleaning. If ferrous shavings are easily removed during cleaning, the cylinder block is not magnetized.



Sources for magnetism:

- Electrically actuated components (such as a clutch).
- Static current from belts or other moving parts.
- Grounding of the electrical system through the crankshaft when some component such as the generator or engine block has **not** been grounded properly.
- An improperly grounded voltage.

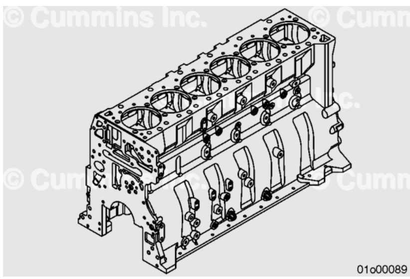
Measure the cylinder block with a gauss meter and record the results. If the magnetism is out of specification, the engine **must** be treated, in general, as if:

- Debris (fine particles) has been traveling throughout the lubrication system, resulting in wear and damage to components

Components with measured gauss units greater than specification **must** be demagnetized or replaced. A facility capable of magnetic testing (Magnaflux) of the engine components is capable of demagnetizing (degaussing) components

Acceptable Magnetism Levels	
Injectors and Ferrous Injector Components	**All Other Ferrous Components**
5 or less gauss units	15 or less gauss units

Before cleaning or further disassembly of the cylinder block, perform an inspection to see if there is any damage (cracks, fretting, etc.) that would prohibit reuse. If any cracks are found other than in the counterbore ledge area, the cylinder block **must** be replaced.



Pay close attention to areas of the cylinder block including:

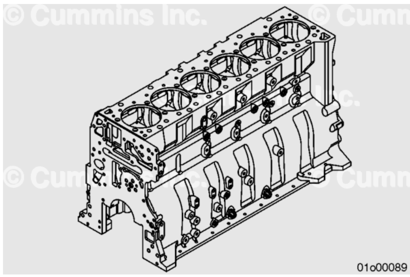
- Main bearing caps and bores
- Cylinder bores
- Cylinder block combustion deck
- Oil pan mounting surface
- Front and rear block sealing surfaces
- Lubricating oil cooler cavity
- Cup plug bores
- Threaded bosses.

Preparatory Steps

Disassemble the engine down to the cylinder block component level.

Reference Section DS - Engine Disassembly - Group 00.

Note : Replace the lubricating oil cooler if the engine has had a debris-generating malfunction.



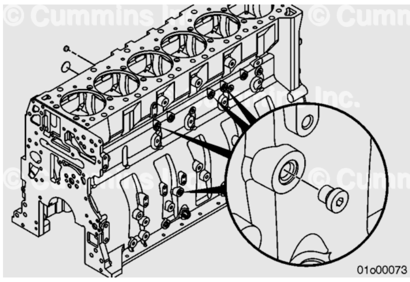
Remove all oil and water gallery cup plugs, threaded plugs, and pipe plugs to allow the cleaning solution to enter the passages.

Remove all cup plugs. Refer to Procedure 017-002 in Section 17. (/qs3/pubsys2/xml/en/procedures/10/10-017-002.html)

Remove all pipe plugs. Refer to Procedure 017-007 in Section 17. (/qs3/pubsys2/xml/en/procedures/99/99-017-007.html)

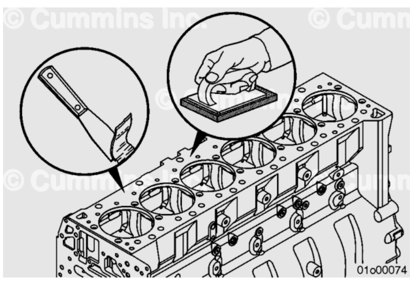
Remove all straight threaded plugs. Refer to Procedure 017-011 in Section 17. (/qs3/pubsys2/xml/en/procedures/10/10-017-011.html)

Note : All cup plugs must be replaced if the component is cleaned in a hot tank, spray washer, or similar equipment. This cleaning activity can interrupt the sealing capacity of the sealant.



⚠ WARNING ⚠

Use caution while handling and cleaning the cylinder block. The cylinder block may contain sharp edges that can cause personal injury.



⚠ CAUTION ⚠

Use care not to damage the machined gasket surfaces.

Use a gasket scraper to clean the cylinder block deck surface.

Use Scotch-Brite™ 7448 abrasive pad, or equivalent, and solvent to remove any residual gasket material from the cylinder block deck surface.

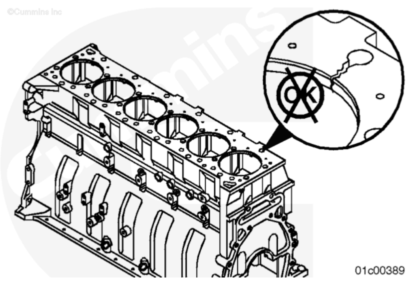
Remove any remaining adhesive sealant from other cylinder block sealing surfaces.

Inspect for Reuse

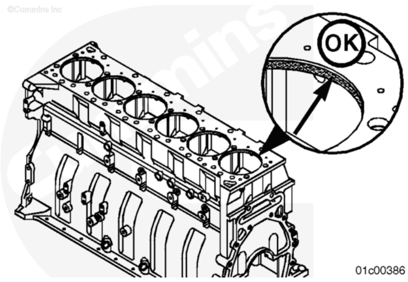
Inspect the cylinder block for cracks. If any cracks are found, other than in the counterbore ledge area, the cylinder block **must** be replaced.

Cracks discovered in the counterbore ledge area **must** be machined and inspected.

Refer to Procedure 001-058 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)



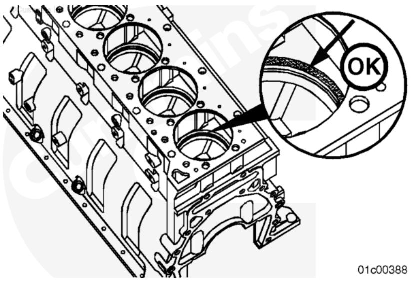
If upper bore erosion is discovered during block inspection, it **must** be disregarded. It is **not** a failure mode and the head gasket provides an adequate seal.



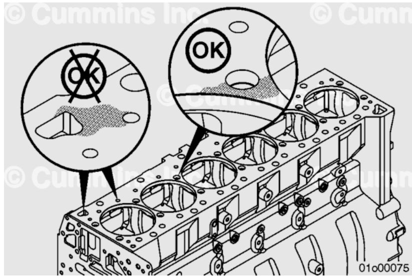
Refer to the Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753 (/qs3/pubsys2/xml/en/bulletin/4383753.html), to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.

If liner ledge pitting, fretting, or wear, is discovered during block inspection, the block **must** be machined.

Refer to Procedure 001-058 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)



Check the top surface of the block for wear.

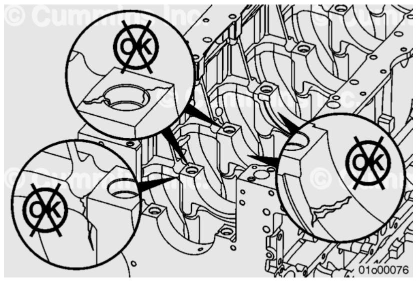


Fretting or pitting of the cylinder block head deck is **not** uncommon. It is **only** detrimental to the head gasket joint if the fretting or pitting is close to the coolant or oil passages. These areas, if damaged, prevent the head gasket from sealing.

If fretting damage, scratches, cracks, or corrosion deeper than 0.08 mm [0.003 in] is present in an area where a head gasket seal ring or a grommet makes contact, the cylinder block **must** be replaced. There **must not** be any damage which extends more than 2.41 mm [0.095 in] from the edge of the coolant passage.

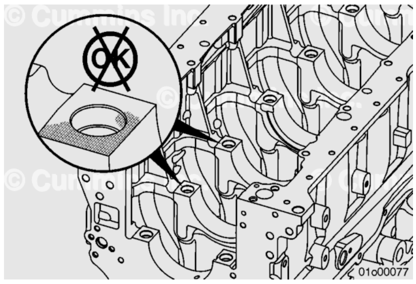
Fretting damage in any other area is acceptable if it does **not** change the protrusion measurement of the cylinder liner.

Inspect the main bearing caps and main bearing saddle areas for cracks or fretting.



If any cracks or fretting are found, the cylinder block **must** be replaced.

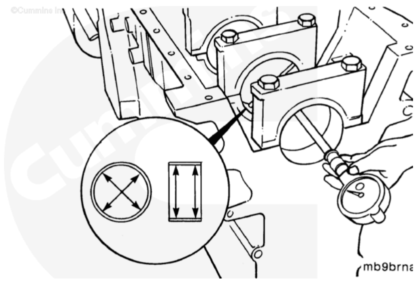
Inspect the main saddles and main bearing caps mating surface for fretting.



Fretting appears as a frosted surface or orange peel texture.

If any fretting is found, the cylinder block **must** be replaced.

The main bearing bore diameter **must** be measured with the main bearings removed and the main bearing caps assembled and tightened to the correct specification.



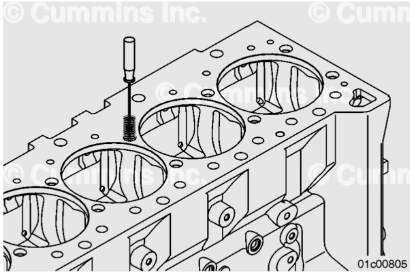
Refer to Procedure 001-006 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-006-tr.html)

Main Bearing Bore Diameter		
mm		in
133.49	MIN	5.2555
133.52	MAX	5.2565

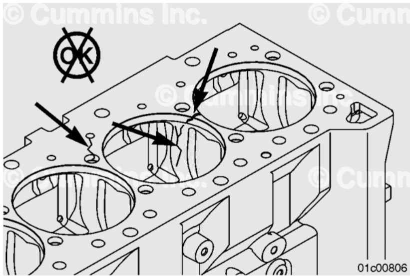
If the main bearing bore diameter does **not** meet specification then the cylinder block **must** be replaced.

Repair

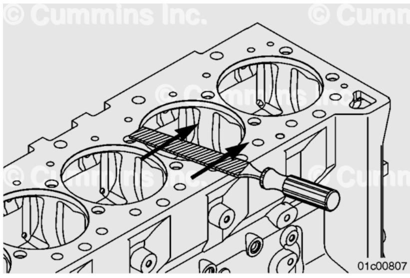
Follow this procedure to salvage capscrew holes that are missing a portion of the threads.



Cracks that extend into the capscrew bosses or the coolant cavity, below the upper counterbore, can **not** be salvaged using this procedure.

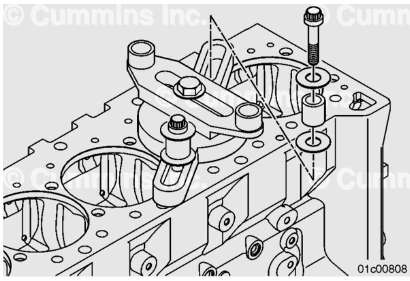


Remove any burrs from the top deck surface of the cylinder block. A 12-inch, flat surface, mill file is effective for this step. Burr removal is necessary to obtain an accurate fixture location.



Note : The thread salvage kit is designed to be used with the cylinder liner removed. If the cylinder liner remains in the block, shims must be installed to both sides of the base equally to bridge the liner.

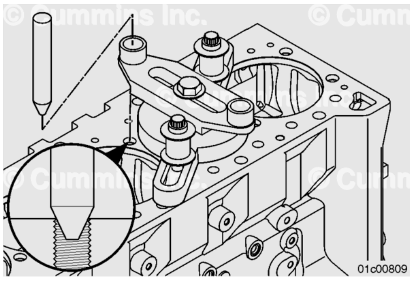
Note : On some engines, it will be necessary to use the spacer, Cummins® Part Number 3376206, between the mounting base and drilling fixture.



Select two capscrew holes so the damaged capscrew hole is positioned approximately in the middle. Install the mounting plate assembly onto the capscrew holes using the appropriate spacers, four plain washers, and the existing cylinder head capscrews.

Use a heavy washer and the 5/8-18 x 3-inch hex head capscrew to attach the drilling fixture to the mounting plate assembly.

Slide the locating pin through the appropriate drilling fixture bushing hole until it rests in the damaged capscrew hole of the cylinder block.



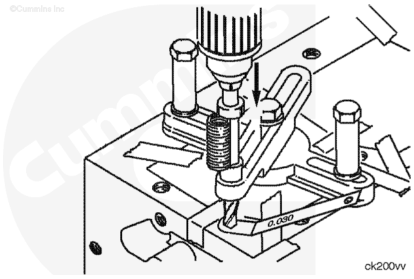
Center the drilling fixture guide bar and bushing over the damaged capscrew hole.

Tighten the three capscrews to secure the fixture assembly.

Torque Value: 68 n•m [50 ft-lb]

Remove the locating pin.

Setting the stop collar is required **only** for repairs using drills.



Drills are used in some kits to allow the quick removal of waste material; however, care **must** be taken to make certain the depth of the repaired capscrew hole is **not** increased beyond specification.

The following steps will result in the drill stopping short of the original hole depth. The remaining material is then removed with the reamer.

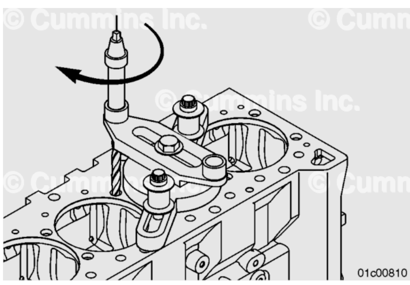
Place a 0.76 mm [0.030 in] feeler gauge over the hole to be repaired.

Insert the drill through the drill bushing until its tip rests against the feeler gauge.

Place the thread insert to be installed between the drill bushing and the stop collar.

Move the stop collar down until it rests against the insert and tighten the stop collar.

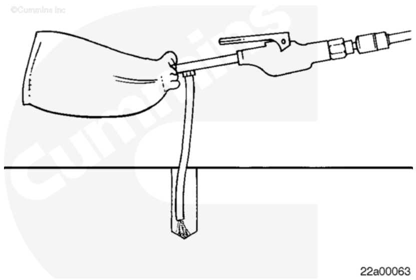
Place the drill through the drill fixture bushing and let it rest over the top of the damaged capscrew hole.



Use a drill motor with a clutch and a maximum operating speed of 300 rpm. Use a suitable cutting fluid. Operate the drill motor until the stop collar contacts the drill bushing.

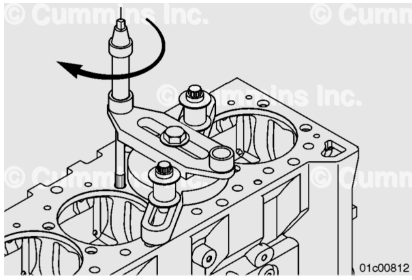
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Remove the chips with a chip vacuum.

Place the reamer through the drill fixture bushing and let it rest over the top of the damaged capscrew hole.



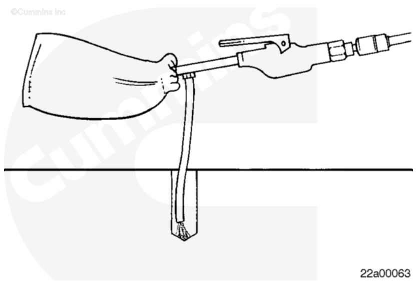
Stop frequently while reaming the hole to clean the hole with the chip vacuum.

Use a drill motor with a clutch and a maximum operating speed of 300 rpm and a suitable cutting fluid. Operate the drill motor until the reamer reaches the bottom of the original capscrew hole.

The reamer will stop cutting when it reaches the bottom of the hole.

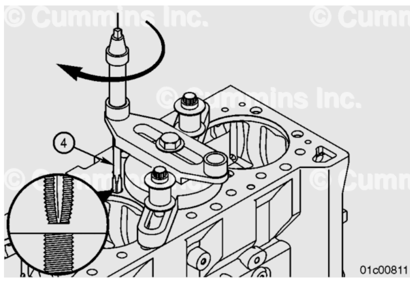
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Remove the chips with a chip vacuum.

Stop the tapping operation frequently to clean the bore using a chip vacuum.



Install a ½-inch square drive socket on the end of the appropriate tap.

Keep the tap well lubricated during tapping.

Install the tap (4). Use a suitable tapping fluid and hand-tap to the bottom of the damaged capscrew hole.

Remove the tap.

⚠ WARNING ⚠

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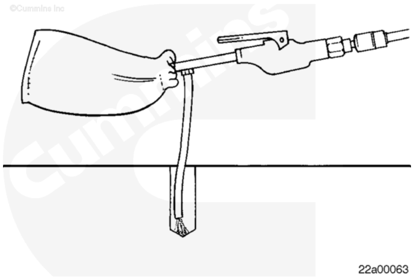
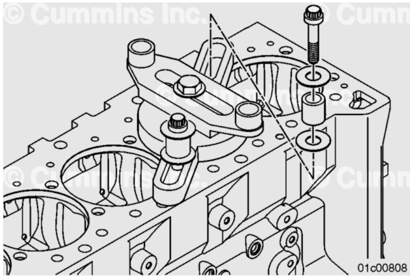
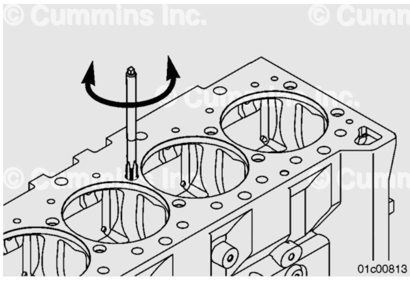
Remove the chips with a chip vacuum.



Remove the mounting capscrews, spacers, four plain washers, and mounting plate assembly.

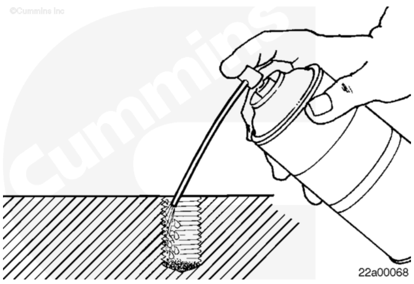


Thread the tap in and out of the newly-cut threads several times to verify the thread condition and clean debris from the threads.



⚠ **WARNING** ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ **WARNING** ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

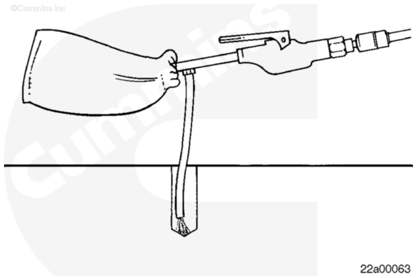
Check the capscrew hole for evidence of porosity and cracks.

Use a solvent-degreasing agent to clean the threads and flush any debris from the tapped hole.

⚠ **WARNING** ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

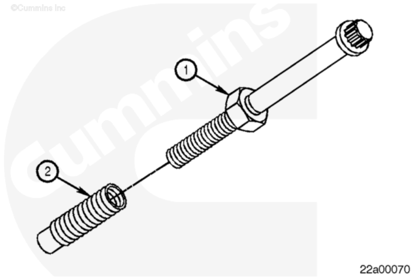
Remove the solvent-degreasing agent from the hole with a chip vacuum.



Place the appropriate jam nut (1) on an appropriate capscrew.

Install the appropriate thread insert (2) onto the cylinder head mounting capscrew until it nears the bottom of the insert.

Rotate the jam nut (1) until it contacts the thread insert (2) to lock the insert and jam nut together.



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⚠ WARNING ⚠

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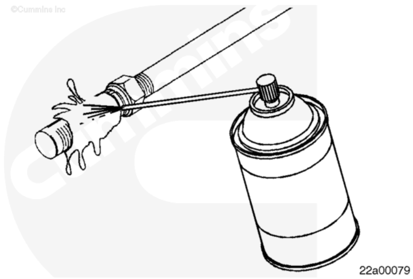
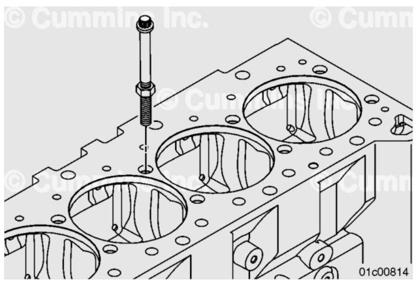
Use a solvent-degreasing agent to remove the preservative coating from the circumference of the thread insert.

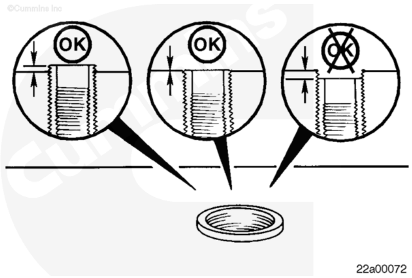


Install the thread insert, jam nut, and capscrew into the newly-tapped hole. Make certain that the thread insert threads freely into the hole.



Note : For applications with a counterbore, measure the protrusion depth starting at the counterbore face, which is recessed into the hole.

Check for protrusion. Make certain the thread insert protrusion is 0 to 1.0 mm [0 to 0.040 in] above the cylinder block surface.



It is important to make certain the minimum required thread depth is maintained after thread repair.

If the insert protrusion exceeds 1 mm [0.040 in], the cylinder head capscrew can bottom out when tightened.

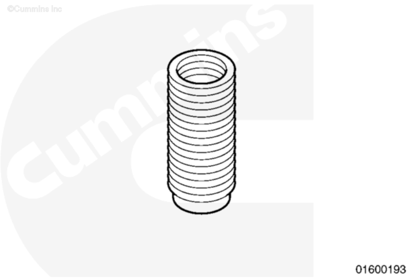
Record the amount the insert protrudes.

It is necessary to increase the depth of the original capscrew hole.

If the insert protrusion is correct, install the insert.

If the protrusion is **not** correct, record the measurement and adjust the bore depth.

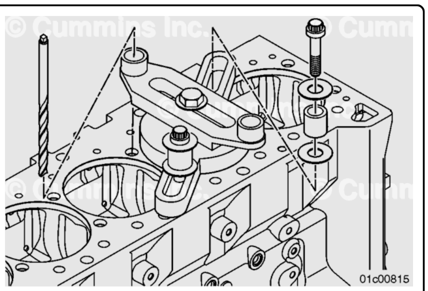
Prior to increasing the capscrew hole depth, check the following:



1. Make sure the correct thread insert is being used. Two inserts are available. The two inserts are identical except for their length.
2. Make certain the insert used is the correct length.
3. Make certain the capscrew hole is threaded all the way to the bottom.
4. Make certain there are no burrs or other damage that would prevent the insert from threading to the bottom of the capscrew hole.

After checking the above listed items, if the thread insert protrusion still exceeds 1.0 mm [0.040 in], the depth of the original capscrew hole **must** be increased.

Note : The thread salvage kit is designed to be used with the cylinder liner removed. If the cylinder liner remains in the block, shims must be installed to both sides of the base equally to bridge the liner.



Note : On some engines, it will be necessary to use the spacer, Part Number 3376206, between the mounting base and drilling fixture.

Install the mounting plate assembly.

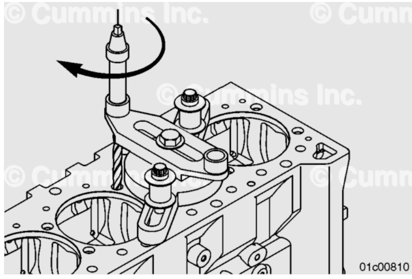
Do **not** tighten the fasteners.

Slide the drill through the drill bushing and into the threaded hole until it rests at the bottom of the hole.

Tighten all fasteners.

Torque Value: 68 n•m [50 ft-lb]

The drill **must** slide freely in the bushing and threaded hole, indicating that it is centered in the hole.



Do **not** increase the depth of the original capscrew hole more than 3.0 mm [0.120 in].

Slide the stop collar over the drill bit.

Subtract 0.50 mm [0.020 in] from the insert protrusion.

Select a feeler gauge equivalent to this value.

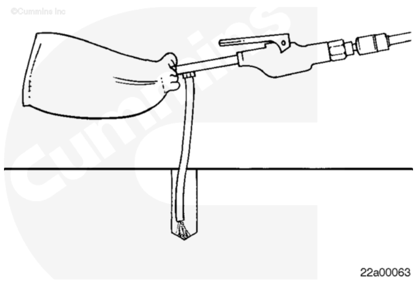
With the drill resting at the bottom of the hole, insert the feeler gauge between the stop collar and drill bushing.

Tighten the stop collar.

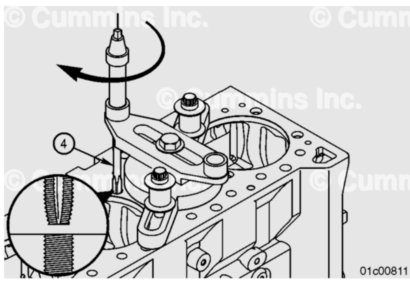
Use a drill motor with a clutch and a maximum operating speed of 300 rpm and a suitable cutting fluid. Operate the drill motor until the stop collar contacts the drill bushing.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Remove the chips with a chip vacuum.



It is **not** necessary to ream the hole prior to the second tapping.

Install a ½ inch square drive socket on the end of the appropriate tap.

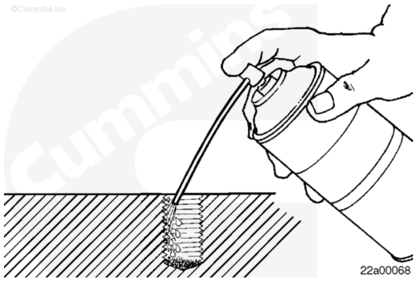
Keep the tap well lubricated during this operation.

Install the tap (4). Use a suitable tapping fluid to hand-tap to the bottom of the damaged capscrew hole.

Remove the tap.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

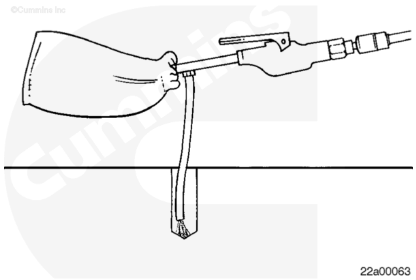
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Check the capscrew hole for evidence of porosity and cracks.

Clean the threads and flush any debris from the tapped hole using a solvent-degreasing agent.

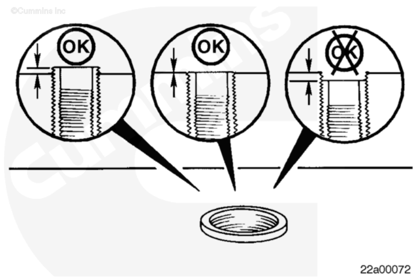
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Remove the chips and solvent-degreasing agent from the hole with a chip vacuum.

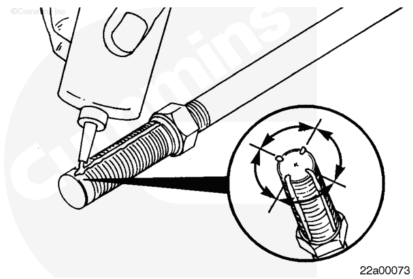
Install the thread insert on a cylinder head capscrew and lock it in place with a jam nut.



Temporarily install the thread insert and check the protrusion. Make certain that the protrusion is 0 to 1 mm [0 to 0.040 in] above the cylinder block surface.

Remove the insert.

Apply a light coat of primer, Part Number 3824715, to the thread insert and the threaded hole. Allow 3 to 5 minutes to dry.

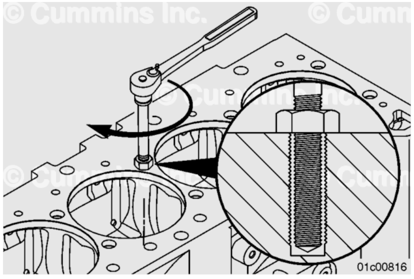


Apply four beads of thread locking compound, Part Number 3824038, to the circumference of the thread insert.

Each bead **must** be 0.8 mm [0.03 in] wide down the entire length of the thread insert and the beads **must** be 90 degrees apart.

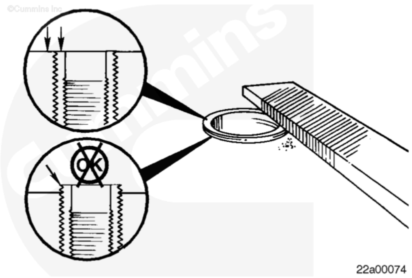
Install the thread insert until it is flush with the cylinder block surface or it bottoms in the hole.

Allow to dry for 3 hours.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

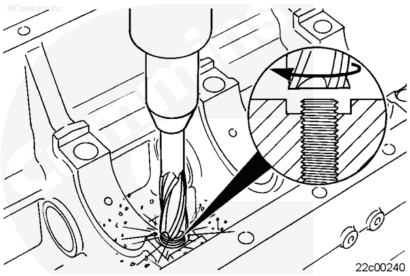


Remove the capscrew and jam nut.

If necessary, file or machine the top of the thread insert so it is flush with the cylinder block surface.

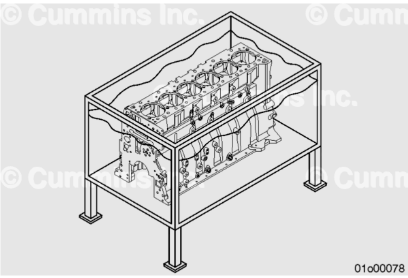
Make certain that the inner diameter of the thread insert is free of burrs. Remove any file shavings or debris with the chip vacuum.

For applications where the thread insert is recessed into a counterbore, an end mill or a machine cutter will be necessary to remove the excess length of the insert and restore the alignment of counterbore.



Clean

Tank Cleaning Method



⚠ WARNING ⚠

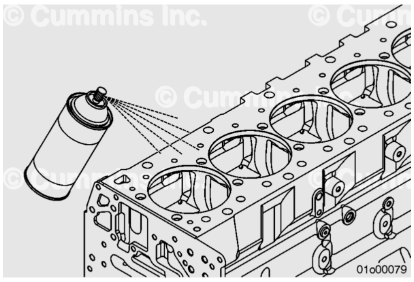
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Cummins Inc. recommends the use of a cleaning tank for a thorough cleaning of the cylinder block. Follow this procedure if a cleaning tank is available.

Note : Cummins Inc. does not recommend any specific cleaning solution.



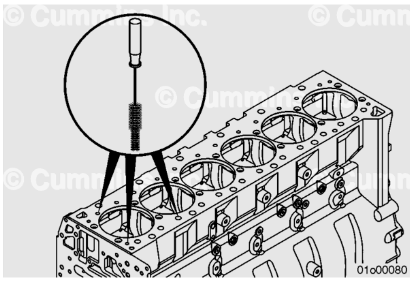
Make sure all oil and water gallery cup plugs, threaded plugs, and pipe plugs are removed to allow the cleaning solution to enter the passages.

Remove all cup plugs. Refer to Procedure 017-002 in Section 17. (/qs3/pubsys2/xml/en/procedures/10/10-017-002.html)

Remove all pipe plugs. Refer to Procedure 017-007 in Section 17. (/qs3/pubsys2/xml/en/procedures/99/99-017-007.html)

Remove all straight threaded plugs. Refer to Procedure 017-011 in Section 17. (/qs3/pubsys2/xml/en/procedures/10/10-017-011.html)

Note : All cup plugs must be replaced if the component is cleaned in a "hot tank", spray washer, or similar equipment. This cleaning activity can interrupt the sealing capacity of the sealant.



⚠ WARNING ⚠

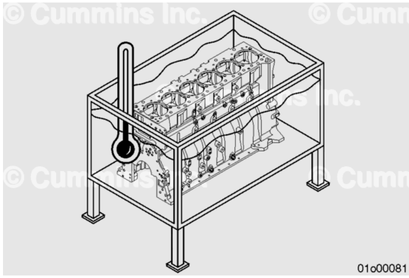
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a gallery brush from an engine cleaning brush kit to clean the oil and coolant galleries. Clean the vertical passages/galleries first, then the horizontal passages/galleries to optimize debris removal.

Use the following procedures for location of the oil and coolant passages. Refer to Procedure 200-002 in Section F. (/qs3/pubsys2/xml/en/procedures/10/10-200-002-tr.html)



⚠ WARNING ⚠

Wear appropriate eye and face protection as well as protective clothing, when using a steam cleaner. Hot steam can cause serious personal injury.

Note : If the cylinder block is not going to the cleaning tank immediately, apply a coating of preservative oil to prevent the formation of rust and cover the cylinder block to prevent dirt from sticking to the oil.

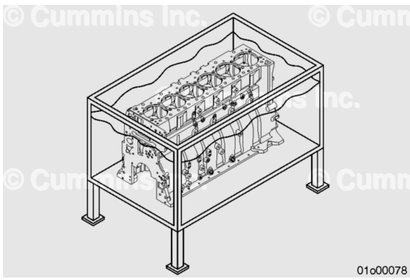
Follow the instructions of the manufacturer of the cleaning tank and the manufacturer of the cleaning solution.

The best results can be obtained by using a cleaning solution that can be heated to 80°C to 95°C [176°F to 203°F].

Use a cleaning tank that will mix and filter the cleaning solution to get the best results.

Remove the cylinder block from the engine stand.

Immerse and agitate the cylinder block in a hot tank bath of cleaning solution.



⚠ WARNING ⚠

Wear appropriate eye and face protection as well as protective clothing, when using a steam cleaner. Hot steam can cause serious personal injury.

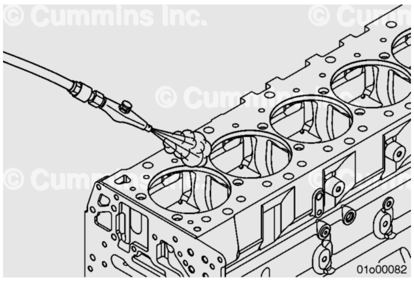
Remove the cylinder block from the cleaning tank.

Wash the cylinder block in hot water or steam clean to remove the cleaning solution.

Rinse the cylinder block in cold water to reduce evaporation and minimize oxidation.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



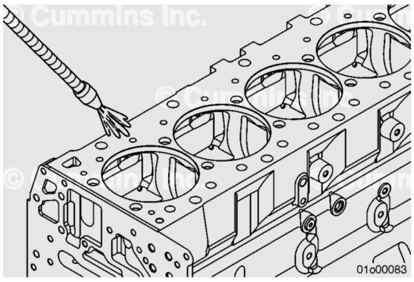
⚠ CAUTION ⚠

Make sure all the water is removed from the capscrew holes and the oil passages to prevent rust formation in the cylinder block.

Dry the cylinder block with compressed air. Blow out all the bolt holes, water and oil passages, or galleries with compressed air.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

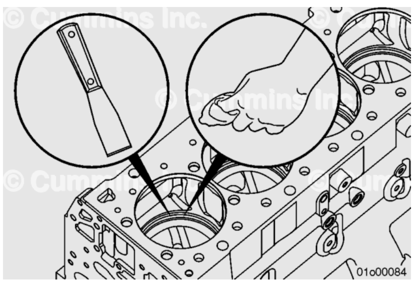


⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

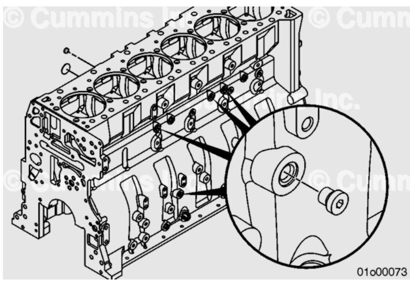
Clean the cylinder head deck surface with safety solvent to remove any residual cleaning solution from the steps above.

Clean the counterbore ledge area and remove any residual cleaning solution, o-ring material, or corrosion.



Clean the cup plug sealing areas.

Note : All cup plugs must be replaced if the component is cleaned in a "hot tank", spray washer, or similar equipment. This cleaning activity can interrupt the sealing capacity of the sealant.



⚠ WARNING ⚠

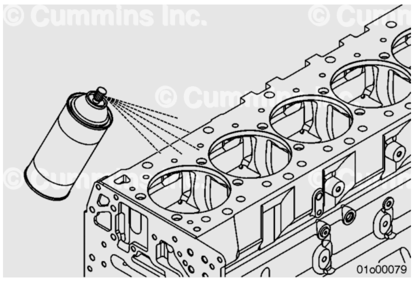
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

If the cylinder block is **not** going to be used immediately, apply a coating of preservative oil to prevent rust.

Cover the cylinder block to prevent dirt from sticking to the oil.



Non-Tank Cleaning Method

Cummins Inc. recommends the use of a cleaning tank for a thorough cleaning of the cylinder block. Follow this procedure if a cleaning tank is **not** available.

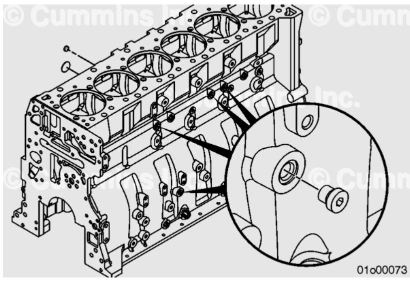
Make sure all oil and water gallery cup plugs, threaded plugs, and pipe plugs are removed to allow cleaning of the passages.

Remove all cup plugs. Refer to Procedure 017-002 in Section 17. (/qs3/pubsys2/xml/en/procedures/10/10-017-002.html)

Remove all pipe plugs. Refer to Procedure 017-007 in Section 17. (/qs3/pubsys2/xml/en/procedures/99/99-017-007.html)

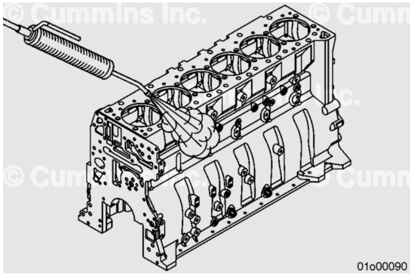
Remove all straight threaded plugs. Refer to Procedure 017-011 in Section 17. (/qs3/pubsys2/xml/en/procedures/10/10-017-011.html)

Note : All cup plugs must be replaced if the component is cleaned in a "hot tank", spray washer, or similar equipment. This cleaning activity can interrupt the sealing capacity of the sealant.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



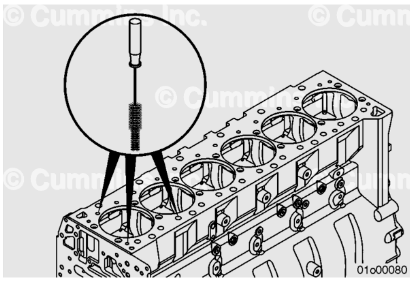
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the cylinder block with a steam cleaner. Make sure all of the oil passages are clean. Use warm, soapy water for cleaning.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



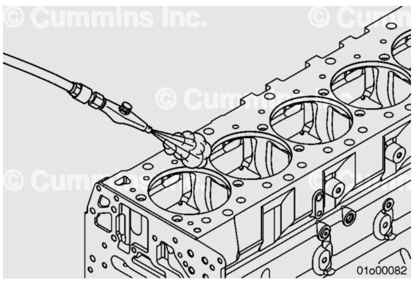
Use a gallery brush from an engine cleaning brush kit to clean the oil and coolant galleries. Clean the vertical passages/galleries first, then the horizontal passages/galleries to optimize debris removal.

Use the following procedures for location of the oil and coolant passages. Refer to Procedure 200-002 in Section F. (/qs3/pubsys2/xml/en/procedures/10/10-200-002-tr.html)

Rinse with cold water to reduce evaporation and minimize oxidation while rinsing.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



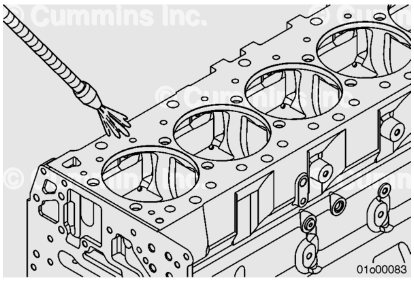
⚠ CAUTION ⚠

Make sure all the water is removed from the capscrew holes and the oil passages to prevent rust formation in the cylinder block.

Dry the cylinder block with compressed air. Blow out all the bolt holes, water and oil passages, or galleries with compressed air.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

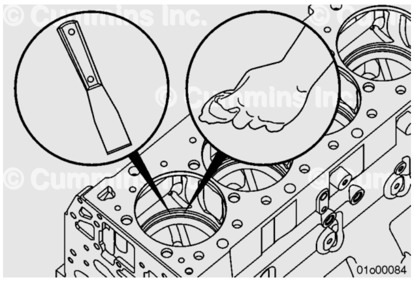


⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

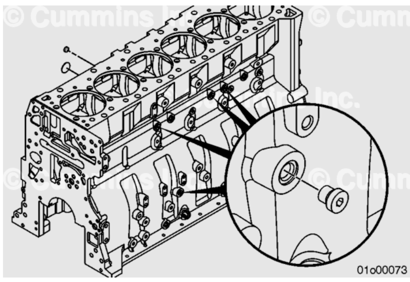
Clean the cylinder head deck surface with safety solvent to remove any residual cleaning solution from the steps above.

Clean the counterbore ledge area and remove any residual cleaning solution, o-ring material, or corrosion.

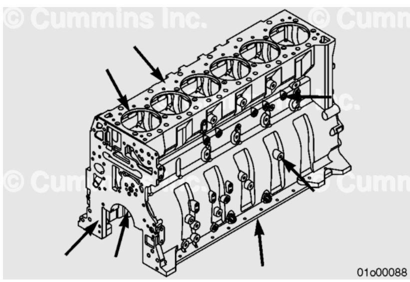


Clean the cup plug sealing areas.

Note : All cup plugs must be replaced if the component is cleaned in a "hot tank", spray washer, or similar equipment. This cleaning activity can interrupt the sealing capacity of the sealant.

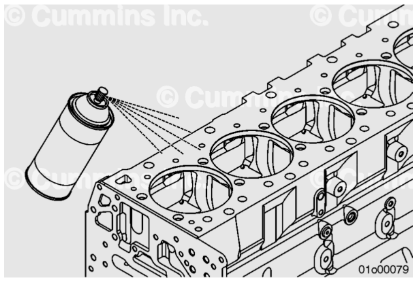


Thoroughly clean all gasket sealing surfaces of any remaining gasket residue.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

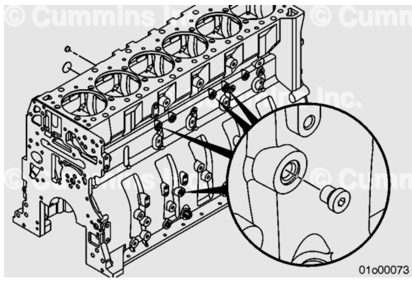
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

If the cylinder block is **not** going to be used immediately, apply a coating of preservative oil to prevent rust.

Cover the cylinder block to prevent dirt from sticking to the oil.

Finishing Steps

Once the cylinder block and all water and oil passages have been thoroughly cleaned and dried:



Install all cup plugs. Refer to Procedure 017-002 in Section 17. (/qs3/pubsys2/xml/en/procedures/10/10-017-002.html)

Install all pipe plugs. Refer to Procedure 017-007 in Section 17. (/qs3/pubsys2/xml/en/procedures/99/99-017-007.html)

Install all straight threaded plugs. Refer to Procedure 017-011 in Section 17. (/qs3/pubsys2/xml/en/procedures/10/10-017-011.html)

Note : All cup plugs must be replaced if the component is cleaned in a "hot tank", spray washer, or similar equipment. This cleaning activity can interrupt the sealing capacity of the sealant.

Note : Before assembling the engine, clean and inspect the crankshaft, camshaft, gear housing, oil pan, cylinder head, rocker levers, lubricating oil pump, pistons, connecting rods, and other associated parts thoroughly to remove debris and replace damaged parts.

Note : If replacing the cylinder block with new, or using a previously stored cylinder block, make sure to clean any oil/rust residue preventative solvent from the cylinder bores, main bearing bores, and gasket sealing areas prior to assembly.

Note : Replace the lubricating oil cooler if the engine has had a debris-generating malfunction.

Assemble the engine.

Reference Section DS - Engine Disassembly - Group 00.